

HUPSEL, Netherlands

WMO: 062830

Lat: 52.0678N

Lon: 6.6567E

Elev: 95

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.7	22.7	12.9	10.6	21.8	17.3	13.2	23.8	26.5	48.0	23.3	46.8	6.2	60	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	85.0	68.0	80.7	66.5	77.1	65.1	70.1	81.0	68.4	77.9	66.6	74.7	7.0	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.6	98.3	74.2	64.9	92.6	72.1	63.5	87.9	70.3	34.1	80.9	32.7	78.3	31.3	75.2	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	17.8	15.7	15.0	93.0	6.3	4.2	10.5	96.0	6.8	98.4	3.3	100.8	-1.3	103.8
			14.2	74.4	6.2	1.9	9.8	75.7	6.2	76.8	2.7	77.9	-1.8	79.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	36.8	38.3	42.7	49.1	55.9	61.2	64.4	63.3	58.3	51.4	44.1	38.4	(6)
(7)		DBStd	11.70	7.73	6.95	6.44	6.57	6.58	5.69	5.71	5.02	5.24	6.26	6.31	7.77	(7)
(8)		HDD50	1711	412	330	236	95	18	0	0	0	2	60	195	363	(8)
(9)		HDD65	5485	876	748	690	476	290	143	82	93	211	423	628	825	(9)
(10)		CDD50	1850	1	2	11	69	202	335	446	413	250	102	16	3	(10)
(11)		CDD65	151	0	0	0	0	9	28	63	41	9	1	0	0	(11)
(12)		CDH74	1567	0	0	0	18	119	314	643	384	85	4	0	0	(12)
(13)		CDH80	443	0	0	0	0	22	81	212	114	14	0	0	0	(13)
(14)	Wind	WSAvg	7.5	9.2	8.9	8.4	7.5	7.3	6.6	6.4	5.8	5.9	7.2	7.7	8.9	(14)
(15)	Precipitation	PrecAvg	33.2	2.9	2.4	2.2	1.8	2.4	2.9	3.3	3.5	2.9	2.8	3.0	3.1	(15)
(16)		PrecMax	44.6	4.9	4.8	5.0	4.1	5.5	6.7	7.7	9.7	7.0	7.7	5.2	7.1	(16)
(17)		PrecMin	25.6	0.2	0.7	0.5	0.0	0.9	0.4	1.2	1.1	0.3	0.9	0.4	1.1	(17)
(18)		PrecStd	4.4	1.3	1.2	1.3	0.9	1.1	1.5	1.5	2.1	1.6	1.3	1.2	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	57.1	65.4	77.1	82.8	88.0	92.2	89.7	81.8	71.8	61.5	55.9	(19)
(20)			MCWB	51.5	49.5	53.8	60.8	65.3	69.0	69.6	70.2	67.4	63.1	56.9	52.7	(20)
(21)		2%	DB	51.8	52.9	60.1	70.7	76.8	81.5	85.5	82.7	76.1	66.2	57.9	53.8	(21)
(22)			MCWB	49.5	48.8	51.7	57.5	61.3	66.3	68.2	68.3	65.7	60.5	54.8	51.4	(22)
(23)		5%	DB	49.5	50.7	55.9	65.9	73.3	77.2	80.9	77.9	71.5	62.7	55.4	51.4	(23)
(24)			MCWB	47.4	47.4	49.9	55.2	59.6	65.0	66.9	66.6	63.5	58.4	52.8	49.2	(24)
(25)		10%	DB	47.1	48.2	52.7	61.5	69.3	73.1	76.7	73.9	67.7	60.3	52.9	48.9	(25)
(26)			MCWB	45.2	45.6	48.0	53.4	58.5	62.6	65.3	64.3	61.3	56.9	50.6	46.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.5	51.9	55.6	61.6	67.2	71.5	72.5	72.7	69.2	63.8	57.4	53.8	(27)
(28)			MCD B	53.7	54.4	61.8	73.8	78.7	84.8	84.8	85.0	77.5	69.7	59.6	55.0	(28)
(29)		2%	WB	50.0	49.6	52.9	58.5	64.0	68.6	70.2	69.7	66.7	61.1	55.1	51.8	(29)
(30)			MCD B	51.5	52.1	58.0	68.0	73.4	78.3	81.6	79.2	74.1	65.4	57.3	53.8	(30)
(31)		5%	WB	47.5	47.6	50.7	56.2	61.5	66.3	68.4	67.6	64.5	59.1	53.1	49.5	(31)
(32)			MCD B	49.3	50.2	55.1	64.1	70.1	74.4	78.4	76.3	70.5	62.2	55.3	51.3	(32)
(33)		10%	WB	45.3	45.5	48.6	54.1	59.4	64.0	66.5	65.6	62.3	57.4	50.9	47.1	(33)
(34)			MCD B	47.0	48.2	52.3	60.6	67.3	71.4	74.8	72.1	66.8	60.0	52.7	48.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.3	10.4	14.0	18.3	18.3	18.3	17.8	17.5	16.3	12.8	9.5	8.0	(35)
(36)			MCDBR	8.7	13.4	20.4	25.6	25.0	25.3	26.4	24.8	21.7	15.8	10.8	8.8	(36)
(37)			MCWBR	7.9	9.9	13.2	14.6	12.3	13.0	11.8	12.5	12.4	10.7	8.7	7.7	(37)
(38)		5% WB	MCDBR	8.9	11.6	17.7	22.7	21.7	21.8	22.6	22.1	19.7	14.8	10.2	8.6	(38)
(39)			MCWBR	8.4	9.4	12.2	13.4	11.7	12.3	11.2	12.0	11.9	10.5	8.7	7.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.327	0.349	0.378	0.410	0.413	0.412	0.423	0.415	0.390	0.363	0.341	0.322	(40)	
(41)		taud	2.359	2.338	2.267	2.216	2.244	2.266	2.242	2.286	2.345	2.402	2.371	2.340	(41)	
(42)		Ebn at Noon	212	241	256	261	266	267	262	257	249	233	203	190	(42)	
(43)		Edh at Noon	20	26	34	40	41	41	41	37	31	24	19	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	221	434	790	1278	1556	1657	1585	1337	973	570	264	171	(44)	
(45)		RadStd	29	79	112	140	139	134	201	116	113	54	31	20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (80 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page